



Cafe Shop Management System

Software Requirement Specification

Project Code: CSMS

Document Code: CSMS-SRS – v1.0

Table of Contents

I. Record of Changes.....	3
II. Software Requirement Specification.....	4
1. Product Overview.....	4
2. User Requirements.....	5
2.1 Actors.....	5
2.2 Use Cases.....	6
3. Software Features.....	10
3.1 Functional Overview.....	10
3.2 Authentication.....	20
3.3 User Administration.....	24
UC02 — Add User.....	26
3.4 Payment.....	28
4. Non-Functional Requirements.....	30
4.1 External Interfaces.....	30
4.2 Quality Attributes.....	30
5. Requirement Appendix.....	31
5.1 Business Rules.....	31
5.2 Common Requirements.....	32
5.3 Application Messages List.....	32
5.4 Other Requirements.....	32

I. Record of Changes

Date	A* M, D	In charge	Change Description
12/06/2026	A	Project Team	Initial version of the SRS for the Cafe Shop Management System.

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Product Overview

The Cafe Shop Management System (CSMS) is a software application designed to support the daily operations of a coffee shop, including menu management, order taking, billing and payment, inventory tracking, table management, and sales reporting. It replaces manual, paper-based processes with a centralized digital platform that improves order accuracy, speeds up service, and gives the owner clear visibility into sales and stock.

The system is used by cafe sales agent at the point of sale and by management for back-office tasks. The context diagram below illustrates the boundary of the system, the external entities (actors and external systems) that interact with it, and the main information flows between them.

Figure 1-1 below presents the context diagram for release 1.0 of the Cafe Shop Management System.

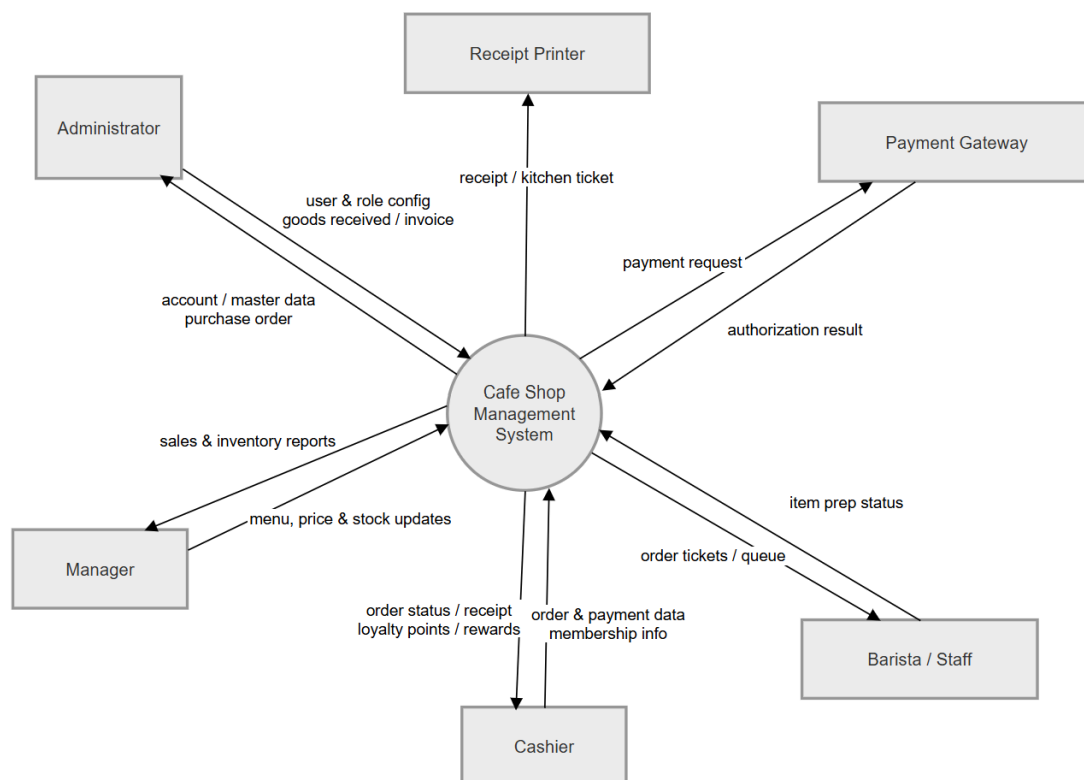


Figure 1-1: Cafe Shop Management System – Context Diagram

As shown in Figure 1-1, the system interfaces with the following external entities:

- **Payment Gateway:** an external service that authorizes and processes card and e-wallet payments submitted by the cashier.

- **Receipt Printer:** a hardware device that prints customer receipts and kitchen order tickets generated by the system.

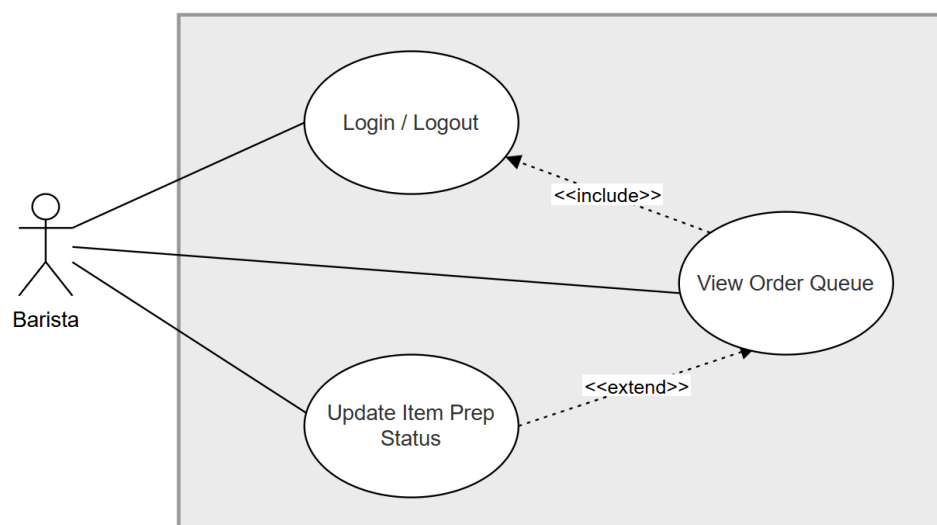
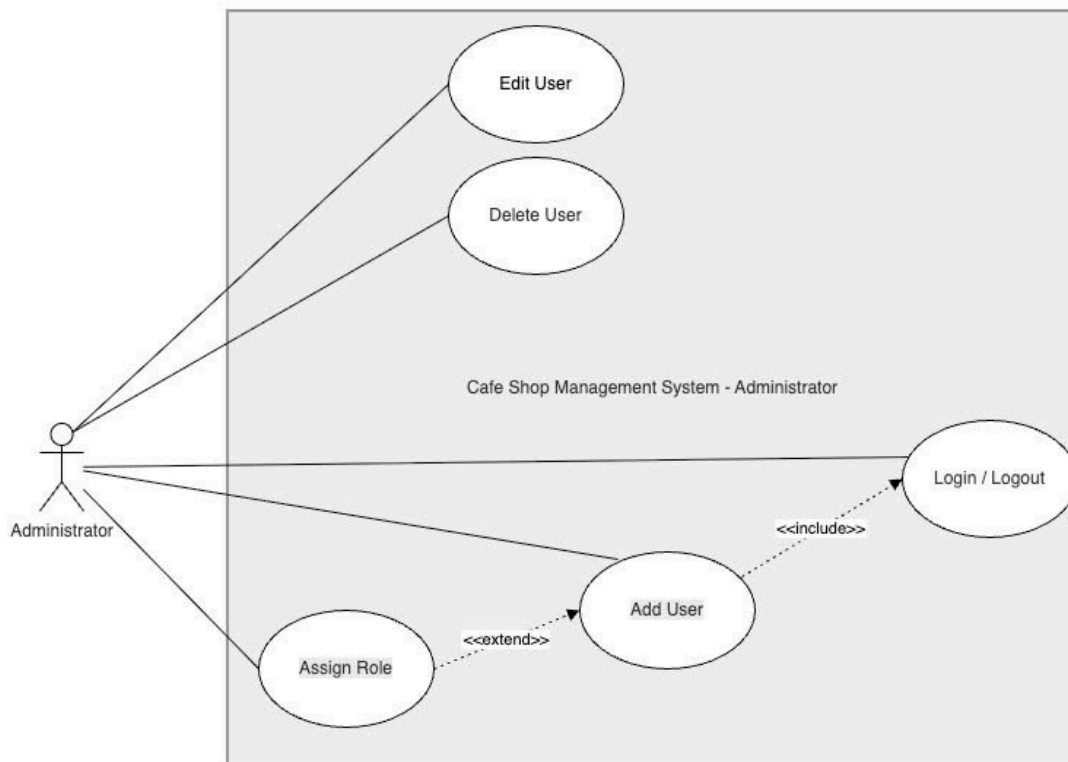
2. User Requirements

2.1 Actors

#	Actor	Description
1	Administrator	Configures the system, manages user accounts and roles, and sets up master data such as products, categories, and tables.
2	Manager	Oversees daily operations, manages the menu and pricing, monitors inventory, and views sales and performance reports.
3	Cashier	Creates and edits customer orders, processes payments, applies discounts, and prints receipts at the point of sale.
4	Sales agent	Receives order tickets, prepares drinks and food, and updates the preparation status of order items.
5	Payment Gateway	External system that authorizes and settles electronic (card / e-wallet) payments.
6	Receipt Printer	External hardware device that prints customer receipts and kitchen order tickets generated by the system.

2.2 Use Cases

2.2.1 Diagram(s)



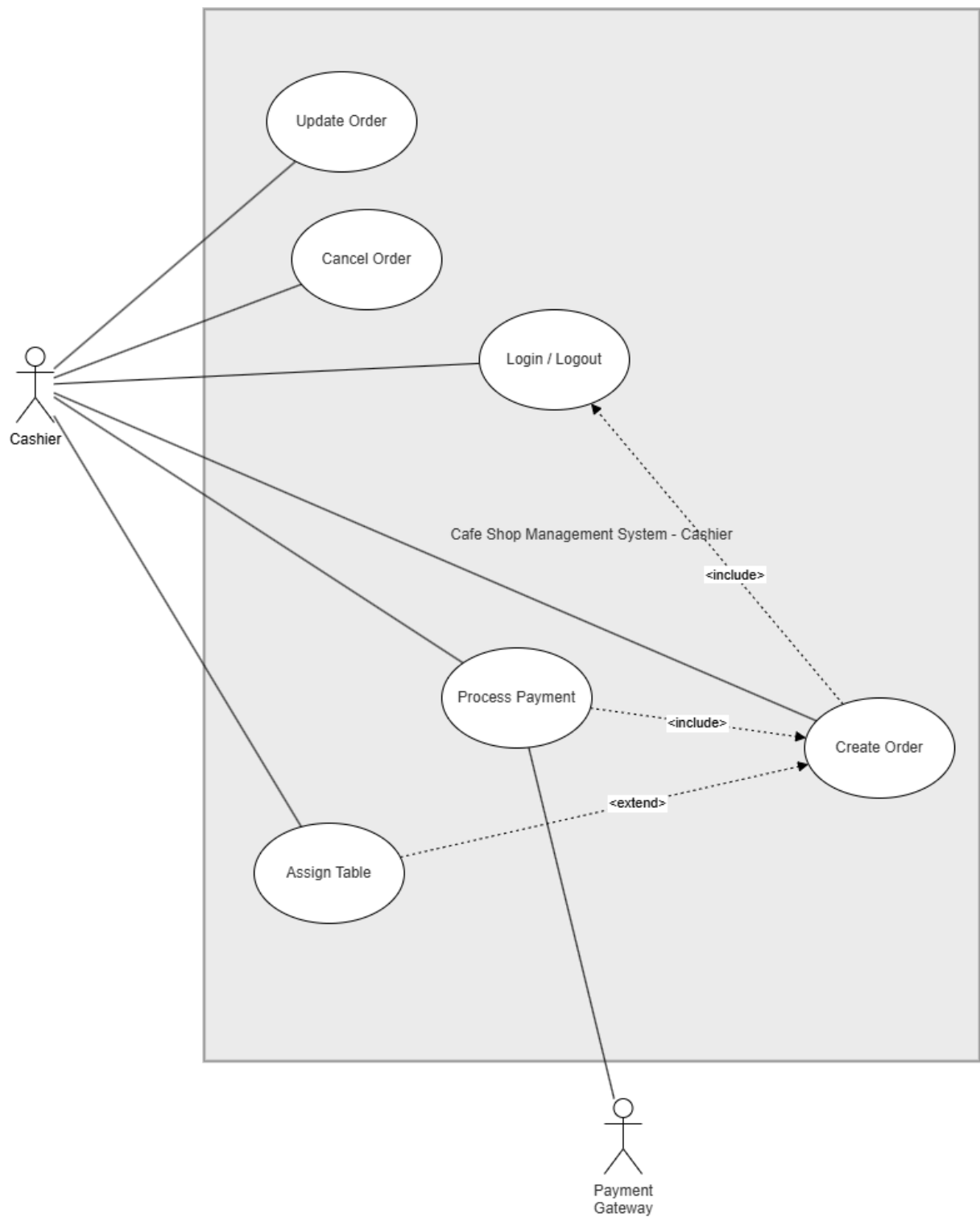




Figure 2-1: Use case diagram of the Cafe Shop Management System

2.2.2 Descriptions

ID	Group function	Use Case	Actors	Use Case Description & Main Flow
01	Authentication	Login / Logout	Administ rator, Manager, Cashier, Barista	A user authenticates with username and password to access role-permitted functions; logout ends the session.
02	Administration	Add User	Administ rator	Creates a new user account.
03	Administration	Edit User	Administ rator	Edits an existing user account.
04	Administration	Delete User	Administ rator	Removes / deactivates a user account.
05	Administration	Assign Role	Administ rator	Assigns a role to a user (extends Add User).
06	Order Management	Create Order	Cashier	Opens a new order, adds items with quantities/options, assigns to a table or takeaway (includes Login).
07	Order Management	Update Order	Cashier	Adds, removes or changes items on an open order before payment.
08	Order Management	Cancel Order	Cashier	Cancels an open order.
09	Order Management	Assign Table	Cashier	Assigns an order to a table (extends Create Order).
10	Payment	Process Payment	Cashier, Payment Gateway	Selects a payment method, applies discount, settles the bill, prints receipt (includes Create Order).
11	Order Queue	View Order Queue	Barista, Cashier, Manager	Views open orders and item preparation status.

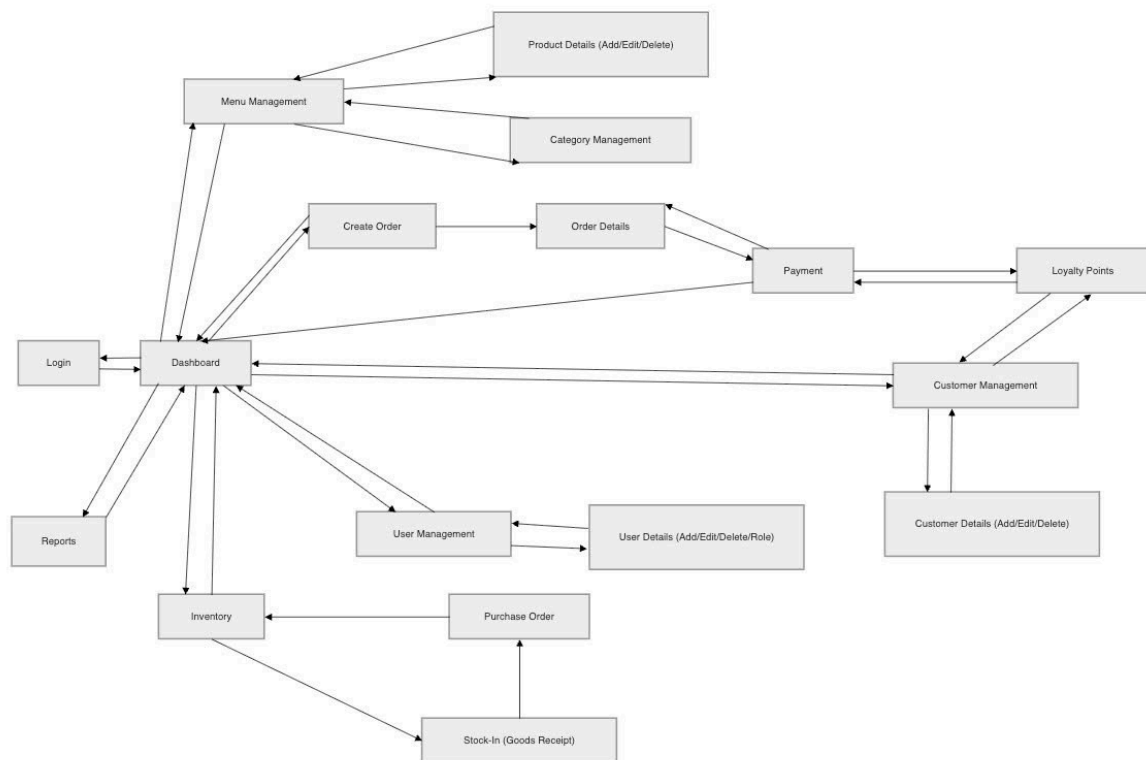
12	Order Queue	Update Item Prep Status	Barista	Marks items in-progress or completed (extends View Order Queue).
13	Menu Management	Add / Edit / Delete / View Product	Manager	Manages products.
14	Menu Management	Add / Edit / Delete Category	Manager	Manages categories (Edit/Delete relate to Add Category).
15	Inventory	Add / Edit / Delete Ingredient	Manager	Manages ingredients.
16	Inventory	Create Stock-In (Goods Receipt)	Manager	Records received goods against a purchase order.

17	Inventory	View Inventory Report	Manager	Reviews stock levels and low-stock items.
18	Table Management	Add / Edit / Delete Table	Manager	Manages tables.
19	Customer	Add / Edit / Delete / View Customer	Manager	Manages customer records.
20	Reporting	View Sales Report	Manager	Views sales/revenue reports by date range.
21	Reporting	Export Report	Manager	Exports a report (includes View Sales Report).

3. Software Features

3.1 Functional Overview

3.1.1 Screens Flow



3.1.2 Screen Descriptions

#	Feature	Screen	Description
1	Authentication	Login	Allows a user to sign in with username and password.

2	Common	Dashboard	Home screen after login; shortcuts to role-permitted features and summary widgets (today's revenue, open orders, low-stock alerts).
3	Order Management	Create Order	Start a new order, add menu items with quantities/options, assign to a table or takeaway.
4	Order Management	Order Details	Shows items, quantities and total; allows editing items before payment.
5	Order Management	View Order Queue	Lists open orders and preparation status; lets baristas mark items in-progress/completed.
6	Payment	Payment	Process the bill, select payment method, apply discount, earn/redeem customer loyalty points, and print receipt.
7	Menu Management	Menu Management	Entry point to manage products and categories.
8	Menu Management	Product Details (Add/Edit/Delete)	Manages individual products.
9	Menu Management	Category Management	Manages categories.
10	Inventory	Inventory	Views stock levels and low-stock alerts.
11	Inventory	Purchase Order	Creates/views purchase orders to suppliers.
12	Inventory	Stock-In (Goods Receipt)	Records received goods against a purchase order.
13	Reporting	Reports	Sales and revenue reports filtered by date range.
14	Administration	User Management	Manages user accounts.
15	Administration	User Details (Add/Edit/Delete/Role)	Manages individual users and roles.
16	Customer	Customer Management	Entry point to manage customers and loyalty.
17	Customer	Customer Details (Add/Edit/Delete)	Manages individual customers.

3.1.3 Screen Authorization

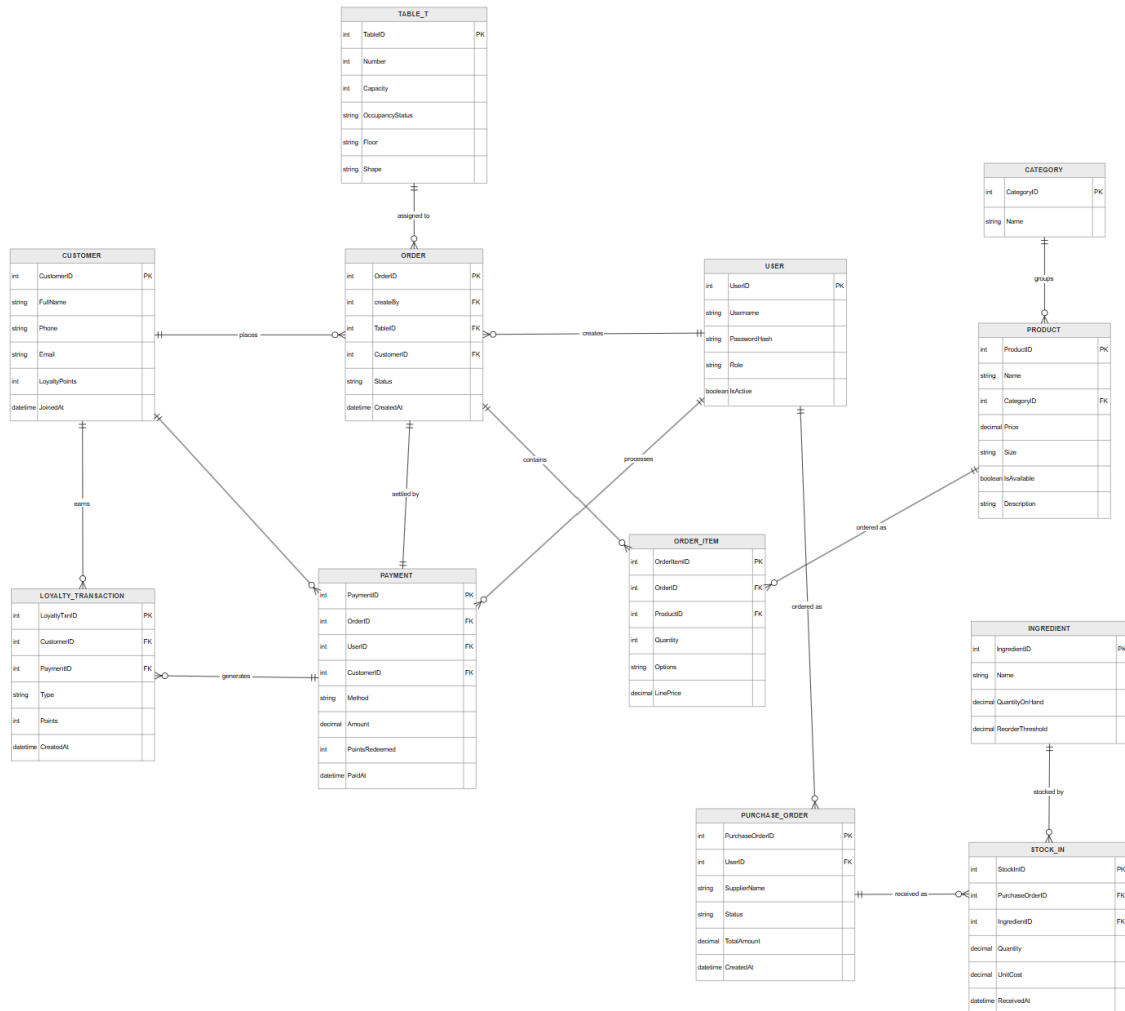
#	Screen	Administrator	Manager	Cashier	Barista
1	Login / Logout	X	X	X	X
2	Dashboard	X	X	X	X
3	Create Order			X	

4	Order Details			X	
5	View Order Queue		X	X	X
6	Payment			X	
7	Menu Management		X		
7.1 Product Details (Add/Edit/Delete)			X		
7.2 Category Management			X		
8	Inventory		X		
8.1 Purchase Order			X		
8.2 Stock-In (Goods Receipt)			X		
9	Reports		X		
10	User Management	X			
10.1 User Details (Add/Edit/Delete/Role)		X			
11	Customer Management		X		
11.1 Customer Details (Add/Edit/Delete)			X		

3.1.4 Non-Screen Functions

#	Feature	System Function	Description
1	Reporting	Daily Sales Aggregation	Scheduled end-of-day job aggregating sales into reporting tables.
2	Inventory	Low-Stock Alert	Background service notifying the manager when an ingredient falls below its reorder threshold.
3	Inventory	Stock Auto-Deduction	On payment, deducts ingredient stock for products in the paid order (BR-08).
4	Payment	Payment Gateway Integration	API integration submitting electronic payment requests and recording the authorization result.
5	Loyalty	Loyalty Points Accrual / Redemption	Within the Payment flow, awards loyalty points to the customer when a payment completes and applies any redeemed points.

3.1.5 Entity Relationship Diagram



Entities Description

#	Entity	Description
1	User	A system user (admin, manager, cashier, barista) with login credentials and a role.
2	Category	A grouping of products (Coffee, Tea, Pastry...).
3	Product	A menu item with name, category, price and availability.

4	Order	A customer order with status, table, customer, creator and timestamp.
5	Order Item	A line in an order: product, quantity, options, line price.
6	Payment	Payment for an order: method, amount, points redeemed, timestamp.
7	Table (TABLE_T)	A physical table with number, capacity and occupancy status.
8	Ingredient	A stock item with quantity on hand and reorder threshold.
9	Customer	A customer with contact info and loyalty points.
10	Loyalty Transaction	A record of points earned/redeemed, linked to a customer and payment.
11	Purchase Order	An order to a supplier, created by a user, with status and total.
12	Stock-In	A goods-receipt line recording received ingredient quantity against a purchase order.

3.1.6 Entity Details

1. User

#	Attribute	PK	Type	Mandatory	Description
1	UserID	x	Integer	Yes	Unique identifier.
2	Username		String	Yes	Login name; unique.
3	PasswordHash		String	Yes	Hashed password.
4	Role		String	Yes	Administrator / Manager / Cashier / Barista.
5	IsActive		Boolean	Yes	Whether the account is active.

2. Category

#	Attribute	PK	Type	Mandatory	Description
1	CategoryID	x	Integer	Yes	Unique identifier.
2	Name		String	Yes	Display name.

3. Product

#	Attribute	PK	Type	Mandatory	Description
1	ProductID	x	Integer	Yes	Unique identifier.
2	Name		String	Yes	Display name.
3	CategoryID	FK	Integer	Yes	FK to Category.
4	Price		Decimal	Yes	Unit selling price.
5	Size		String	No	Size variants (S/M/L).
6	IsAvailable		Boolean	Yes	Available for ordering.
7	Description		String	No	Optional description.

4. Order

#	Attribute	PK	Type	Mandatory	Description
1	OrderID	x	Integer	Yes	Unique identifier.
2	createBy	FK	Integer	Yes	FK to User (creator).
3	TableID	FK	Integer	No	FK to Table; null for takeaway.
4	CustomerID	FK	Integer	No	FK to Customer.
5	Status		String	Yes	Open / Paid / Cancelled.
6	CreatedAt		Datetime	Yes	Creation timestamp.

5. Order Item

#	Attribute	PK	Type	Mandatory	Description
1	OrderItemID	x	Integer	Yes	Unique identifier.
2	OrderID	FK	Integer	Yes	FK to Order.
3	ProductID	FK	Integer	Yes	FK to Product.
4	Quantity		Integer	Yes	Number of units.
5	Options		String	No	Modifiers (size, sugar level, notes).
6	LinePrice		Decimal	Yes	Line total.

6. Payment

#	Attribute	PK	Type	Mandatory	Description
1	PaymentID	x	Integer	Yes	Unique identifier.
2	OrderID	FK	Integer	Yes	FK to Order.
3	UserID	FK	Integer	Yes	FK to User (who processed it).
4	CustomerID	FK	Integer	No	FK to Customer.
5	Method		String	Yes	Cash / Card / E-Wallet.
6	Amount		Decimal	Yes	Amount paid.
7	PointsRedeemed		Integer	No	Loyalty points redeemed.
8	PaidAt		Datetime	Yes	Completion timestamp.

7. Table (TABLE_T)

#	Attribute	PK	Type	Mandatory	Description
1	TableID	x	Integer	Yes	Unique identifier.

2	Number		Integer	Yes	Table number.
3	Capacity		Integer	Yes	Number of seats.
4	OccupancyStatus		String	Yes	Free / Occupied / Reserved.
5			String	Yes	
6	Shape		String	Yes	Round/Square/

8. Ingredient

#	Attribute	PK	Type	Mandatory	Description
1	IngredientID	x	Integer	Yes	Unique identifier.
2	Name		String	Yes	Display name.
3	QuantityOnHand		Decimal	Yes	Current stock level.
4	ReorderThreshold		Decimal	Yes	Low-stock alert level.

9. Customer

#	Attribute	PK	Type	Mandatory	Description
1	CustomerID	x	Integer	Yes	Unique identifier.
2	FullName		String	Yes	Customer name.
3	Phone		String	No	Phone number.
4	Email		String	No	Email address.
5	LoyaltyPoints		Integer	Yes	Current loyalty points balance.
6	JoinedAt		Datetime	Yes	Registration timestamp.

10. Loyalty Transaction

#	Attribute	PK	Type	Mandatory	Description
1	LoyaltyTxnID	x	Integer	Yes	Unique identifier.
2	CustomerID	FK	Integer	Yes	FK to Customer.
3	PaymentID	FK	Integer	Yes	FK to Payment.
4	Type		String	Yes	Earn / Redeem.
5	Points		Integer	Yes	Points earned or redeemed.
6	CreatedAt		Datetime	Yes	Timestamp.

11. Purchase Order

#	Attribute	PK	Type	Mandatory	Description
1	PurchaseOrderID	x	Integer	Yes	Unique identifier.
2	UserID	FK	Integer	Yes	FK to User (creator).
3	SupplierName		String	Yes	Supplier name.
4	Status		String	Yes	Order status.
5	TotalAmount		Decimal	Yes	Total cost.
6	CreatedAt		Datetime	Yes	Timestamp.

12. Stock-In

#	Attribute	PK	Type	Mandatory	Description
1	StockInID	x	Integer	Yes	Unique identifier.
2	PurchaseOrderID	FK	Integer	Yes	FK to Purchase Order.
3	IngredientID	FK	Integer	Yes	FK to Ingredient.
4	Quantity		Decimal	Yes	Quantity received.
5	UnitCost		Decimal	Yes	Cost per unit.
6	ReceivedAt		Datetime	Yes	Receipt timestamp.

3.2 Authentication

3.2.1 FUNC-01 – Login / UC01 Login

3.2.1.1 Screen Mock-up

The mock-up shows a mobile application interface for 'Brew & Co. COUNTER POS'. At the top, the status bar shows the time 9:41 and signal/battery icons. The app header features a coffee cup icon and the text 'Brew & Co. COUNTER POS'. The login form consists of a 'Username' field with the text 'cashier.linh' and a 'Password' field with masked characters and a toggle icon. Below the fields is an orange 'Login' button. A link 'Trouble signing in? Ask your manager.' is positioned below the button. The footer displays 'CSMS v2.0 · Store #HCM-01'.

9:41



Brew & Co.
COUNTER POS

Username

 cashier.linh

Password

Login

Trouble signing in? Ask your manager.

CSMS v2.0 · Store #HCM-01

Figure 3-1: Screen Design of Login

Table 3-1: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	User Name	Text	Yes	20	
2	Password	Text	Yes	20	Display "*" instead of clear character
3	Login	Button			Go to home page

3.2.1.2 Use Case Description

Use Case ID	UC10	Use Case Name		Login	
Author	Project Team	Version	1.0	Date	12/06/2026
Actor	Administrator, Manager, Cashier, Sales agent				
Description	This function allows a user to log in to the Cafe Shop Management System using a registered and active account, in order to access the functions permitted by their assigned role.				
Precondition	PRE-01: The user has a registered account in the system. PRE-02: The user's account is active and not locked.				
Trigger	TRG-01: The user opens the Login screen and submits credentials.				
Post-Condition	POS-01: The user is authenticated and the Dashboard is displayed according to the user's role.				
Main flows					
Step	Actor	Action			
1	User	Open the Cafe Shop Management System mobile app.			
2	CSMS	Display Login screen with the following fields:			

		- User Name - Password - Login button
3	User	Enter user name & password into User Name & Password fields on the Login screen, then click on Login button.
4	CSMS	Validate the entered user name and password, then display the Dashboard screen.

3.2.2 FUNC-02 – Logout / UC11 Logout

3.2.2.1 Screen Mock-up

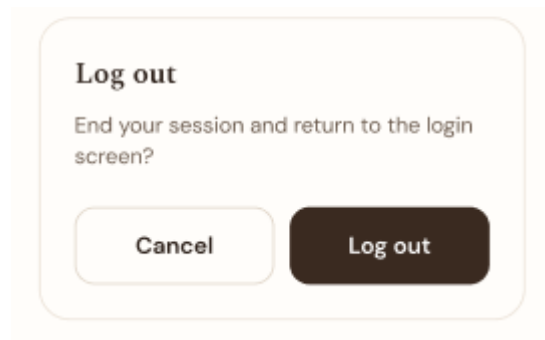


Table 3-2: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Delete Confirmation Message	Label	No	100	Display the message “Are you sure you want to delete this item?”
2	Cancel Delete	Button	No		Close delete confirmation
3	Delete	Button	No		Delete the selected item and close the dialog.
4	Logout Confirmation Message		No	100	Display the message “End your session and return to the login screen?”
5	Cancel Logout	Button	No		Close the logout confirmation

6	Log out	Button	No		Terminate the current session and navigate to the login screen.
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3.2.2.2 Use Case Description

Use Case ID	UC02	Use Case Name		Logout	
Author	Project Team	Version	1.0	Date	12/06/2026
Actor	Administrator, Manager, Cashier, Barista				
Description	A logged-in user ends the session and returns to the Login screen.				
Precondition	The user is logged in.				
Trigger	TRG-01: User selects Logout.				
Post-Condition	POS-01: Session ends; Login screen is displayed.				
Main flows					
Step	Actor	Action			
1	User	Select Logout.			
2	CSMS	End session, clear context, display Login screen.			

3.3 User Administration

3.3.1 FUNC-01/UC04: View User List

3.3.1.1 Screen Mock-Up

9:41



Users

12 accounts · 9 active

Q Search users...

LN

Linh Nguyen
cashier.linh

Cashier

TM

Tran Minh
manager.minh

Manager

PA

Pham An
barista.an

Barista

DM

Do Mai
staff.mai · inactive

Off



#	Field Name	Type	Mandatory	Max Length	Description
1	Back Button	Button	No		Navigate back to the previous dashboard screen.
2	Screen Title	Label	No	50	Display the static screen title “Users”.
3	Accounts Summary	Label	No	100	Display the short summary of total and active accounts (e.g., “12 accounts · 9 active”).
4	Search users Input	Text Input	No	100	Allow searching users by typing keywords or usernames. Placeholder: “Search users...”.
5	User List Container	Container	No		Scrollable area displaying the list card of all accounts in the system.
6	User Avatar Avatar	Image / Icon	No		Circular badge displaying the user's initials (e.g., “LN”, “TM”, “PA”, “DM”) based on their full name. Greyed out if inactive.

7	User Name Label	Label	No	50	Display the full name of the user (e.g., “Linh Nguyen”, “Tran Minh”, “Pham An”, “Do Mai”).
8	Username & Status Text	Label	No	100	Display the unique system handle (e.g., “cashier.linh”, “manager.minh”). Append “ · inactive” if the account is deactivated.
9	Role Badge	Label	No	20	Display a colored chip indicating the assigned system role (e.g., “Cashier”, “Manager”, “Barista”). Displays “Off” in grey if the user is inactive.
10	User Item Row	Container	No		Trigger an event when clicked, navigating the Administrator to the corresponding Edit User screen.
11	Add User Button	Button	No		Floating Action Button (FAB) positioned at the bottom right with a “+” icon to navigate to the Add User screen.

3.3.1.2 Use Case Description

Use Case ID	UC04		Use Case Name		View User List
Author	Project Team	Version	1.0	Date	12/06/2026
Actor		Administrator			
Description		The administrator views the list of all user accounts in the system.			
Precondition		The administrator is logged into the system.			
Trigger		Administrator selects the "User Management" or "User List" navigation option.			
Post-Condition		The system displays the comprehensive list of user accounts.			
Main flows					
Step		Actor		Action	
1		Admin		Selects "User List" from the menu.	
2		CSMS		Retrieves the list of user accounts from the database.	
3		Admin		Displays the user list page with detailed information (e.g., ID, Name, Email, Role, Status).	
Alternative flows					
ID		Action			
AT1		The user list is empty $\rightarrow$ CSMS displays a message indicating that no users were found.			

3.3.2 FUNC-03 / UC02: Add User

3.3.2.1 Screen Mock-up

9:41

×

Add User

Full name

Linh Nguyen

Username

cashier.linh

Role

Administrator

Manager

Cashier

Barista

Account active

Can sign in to CSMS

Add User

Table 3-3: Table Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Close (X)	Button	-	-	Cancels the current operation and navigates back to the User Management list screen.
2	Delete	Icon/Button	-	-	Clears all currently entered data from the form fields.
3	Full name	Text Input	Yes	50	Enters the full name of the employee being granted the account.
4	Username	Text Input	Yes	30	Enters the unique identifier used to log into the system. Displays a user icon at the start of the field.
5	Role	Radio Button Group	Yes	-	Selects a single permission level for the account
6	Account active	Toggle Switch	Yes	-	Sets the activation status of the account
7	Add User	Button	-	-	Validates data entry. Creates the account and triggers a success message upon completion.

3.3.2.2 Use Case Description

UC02 — Add User

Use Case ID	UC02	Use Case Name	Add User
-------------	------	---------------	----------

Author	Project Team	Version	1.0	Date	12/06/2026
Actor		Administrator			
Description		The administrator creates a new user account and assigns a role.			
Precondition		Administrator is logged in.			
Trigger		Administrator opens User Management and selects Add User.			
Post-Condition		A new active user account exists.			
Includes		UC01 (Login)			
Extended by		UC05 (Assign Role)			
Main flows					
Step		Actor		Action	
1		Admin		Open User Management, click Add User.	
2		CSMS		Display the user form (Username, Password, Role, IsActive).	
3		Admin		Enter details and Save.	
4		CSMS		Validate, create the account, show MSG05-style success.	
Alternative flows					
ID		Action			
AT1 / 3.1		Required field empty → MSG02.			
AT2 / 3.2		Field exceeds max length → MSG08.			
AT3 / 4.1		Username already exists → show duplicate-username message.			

3.3.3 FUNC-04 - Edit User

3.3.3.1 Screen Mockup

9:41



Edit User



Full name

Linh Nguyen

Username

 cashier.linh

Role

Administrator



Manager



Cashier



Barista



Account active

Can sign in to CSMS



Save changes

#	Field Name	Type	Mandatory	Max Length	Description
1	Delete Confirmation Message	Label	No	100	Display the message “Are you sure you want to delete this item?”
2	Cancel Delete	Button	No		Close delete confirmation
3	Delete	Button	No		Delete the selected item and close the dialog.
4	Logout Confirmation Message		No	100	Display the message “End your session and return to the login screen?”
5	Cancel Logout	Button	No		Close the logout confirmation
6	Log out	Button	No		Terminate the current session and navigate to the login screen.

3.3.3.2 Use Case Description

Use Case ID	UC03	Use Case Name	Edit User
--------------------	------	----------------------	-----------

Author	Project Team	Version	1.0	Date	12/06/2026
Actor		Administrator			
Description		The administrator edits an existing user account (details, role, active status).			
Precondition		The target user account exists.			
Trigger		Administrator selects a user and chooses Edit.			
Post-Condition		The user account is updated.			
Main flows					
Step		Actor		Action	
1		Admin		Open User Management, select a user, click Edit.	
2		CSMS		Display the form pre-filled with current data.	
3		Admin		Change values and Save.	
4		CSMS		Validate and save the changes.	
Alternative flows					
ID		Action			
AT1 / 3.1		Required field empty → MSG02.			
AT2 / 3.2		Over max length → MSG08.			

3.4 Payment

3.4.1 FUNC-04-Process Payment / UC10: Process Payment

3.4.1.1 Screen Mock-up

9:41

< Payment
#ORD-1182 · T-05

Amount due
205,000đ

Payment method

☒ Cash Card E-Wallet

ML Mai Le
Balance 1,240 pts +20 pts

Redeem 500 pts
-50,000đ off total - 500 +

Subtotal 225,000đ
Discount (loyalty) -50,000đ
Cash tendered 300,000đ

Change due **125,000đ**

Confirm payment · 175,000đ

Table 3-2: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
---	------------	------	-----------	------------	-------------

1	Amount due	Label			Displays the total amount the customer needs to pay.
2	Payment method	Button Group	Yes		Allows selection of payment method: Cash, Card, E-Wallet.
3	Redeem pts	Numeric Input	No		Input field for loyalty points to redeem (includes - / + buttons).
4	Discount	Label			Displays the applied discount percentage and amount.
5	Cash tendered	Numeric Input	Yes		Input field for the amount of cash received (required if Cash is selected).
6	Request approval	Button			Sends an approval request to the Manager (appears if discount > 50%).
7	Confirm payment	Button			Confirms the payment and completes the order.
8	Print receipt	Button			Prints the order receipt (shown on success screen).
9	New order	Button			Navigates back to create a new order (shown on success screen).

3.4.1.2 Use Case Description

Use Case ID	UC10	Use Case Name		Process Payment	
Author	Project Team	Version	1.0	Date	12/06/2026
Actor	Cashier, Payment Gateway				
Description	The cashier settles the bill: selects a method, applies discount, earns/redeems loyalty points, and prints the receipt.				
Precondition	An order with status Open and ≥ 1 item exists.				
Trigger	The cashier selects Payment on an open order.				

Post-Condition		A payment is recorded; order status becomes Paid; receipt printed; stock deducted; loyalty points updated.
Main flows		
Step	Actor	Action
1	Cashier	Select Logout.
2	CSMS	Show total, discount field, method options, loyalty section.
3	Cashier	Apply discount (if any), select method, set points to earn/redeem.
4	CSMS	If electronic, send payment request to the Payment Gateway
5	Payment Gateway	Return authorization result.
6	CSMS	Record payment, set order Paid, deduct stock (BR-08), update loyalty points, print receipt .

3.5 Order Management

3.5.1 FUNC-05-Create Order/ UC6: Create Order

3.5.1.1 Screen Mock-up



Table 3-3: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Back	Button			Returns to the previous screen without saving the order.
2	Cancel	Button			Discards the current new order and exits the screen.
3	Table Selector	Button	No		Opens table selection to assign a table and number of seats (e.g., T-05 · 4 seats).
4	Add Customer	Button	No		Attaches an existing customer to the order for loyalty tracking.
5	Search Menu	Text Input	No	50	Searches the menu for products by name.
6	Category Tabs	Tab / Button	No		Filters products by category (Coffee, Tea, Pastry, Cold).
7	Product Card	Container	No		Displays a product with name, price and size options (e.g.,

					S/M/L); tapping adds it to the order.
8	Order Summary	Label			Shows the running item count and total amount of the order.
9	Review Order	Button			Proceeds to the order review and confirmation step.

3.5.1.2 Use Case Description

Use Case ID	UC06		Use Case Name		Create Order
Author	Project Team	Version	1.0	Date	12/06/2026
Actor		Cashier			
Description		The cashier opens a new order, adds products with quantity and options, and assigns it to a table or takeaway.			
Precondition		The cashier is logged in; products exist and are available.			
Trigger		Cashier selects Create Order.			
Post-Condition		An order with status Open is saved.			
Includes		UC01 (Login)			
Extended by		UC09 (Assign Table)			
Main flows					
Step		Actor		Action	
1		Cashier		Select Create Order.	
2		CSMS		Display menu by category.	
3		Cashier		Add products, set quantity and options (size, sugar, notes).	
4		CSMS		Update the running order summary and total.	
5		Cashier		Save the order.	
6		CSMS		Create the order (status Open), show MSG03.	
Alternative flows					

ID	Action
AT1 / 5.1	Order has no item (BR-01) → block save, prompt to add a product.
AT2 / 3.1	Selected product is unavailable/out of stock (BR-04) → disabled / not addable.
Business Rules	BR-01, BR-02, BR-04

3.5.2 FUNC-05-Order Management/ UC7: Update Order

3.5.2.1 Screen Mock-up

9:41

Coffee Tea

C Cappuccino
45,000đ

Size

S · 40k M · 45k L · 52k

Sugar level

0% 30% 50% 100%

Notes optional

e.g. extra hot, oat milk...

Quantity

– 2 +

Add 2 · 90,000đ

Table 3-4: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Product Name	Label			Displays the selected product name (e.g., Cappuccino).
2	Product Price	Label			Displays the unit price of the product (e.g., 45,000đ).
3	Size	Radio Button Group	Yes		Selects one size option, each with its own price (e.g., S·40k, M·45k, L·52k).
4	Sugar Level	Radio Button Group	Yes		Selects the sugar level (0%, 30%, 50%, 100%).
5	Notes	Text Input	No	200	Optional free-text note for the item (e.g., extra hot, oat milk).
6	Quantity	Numeric Input	Yes		Sets the item quantity using the – / + stepper (minimum 1).
7	Add to Order	Button			Adds the item with the selected options to the order; the button shows the quantity and line total (e.g., Add 2 · 90,000đ).

3.5.2.2 Use Case Description

Use Case ID	UC07		Use Case Name		Update Order
Author	Project Team	Version	1.0	Date	12/06/2026
Actor		Cashier			
Description		The cashier adds, removes or changes items on an open order before payment.			
Precondition		An order with status Open exists.			
Trigger		Cashier opens an open order and edits items.			
Post-Condition		The order is updated; total recalculated.			
Main flows					
Step		Actor		Action	
1		Cashier		Open an existing open order.	
2		CSMS		Display Order Details.	
3		Cashier		Add/remove/change items and quantities.	
4		CSMS		Recalculate the total and save.	
Alternative flows					
ID		Action			
AT1 / 3.1		Order is already Paid (BR-07) → editing blocked.			
Business Rules		BR-01, BR-02, BR-07			

3.5.3 FUNC-05-Order Management/ UC8: Cancel Order

3.5.3.1 Screen Mock-up

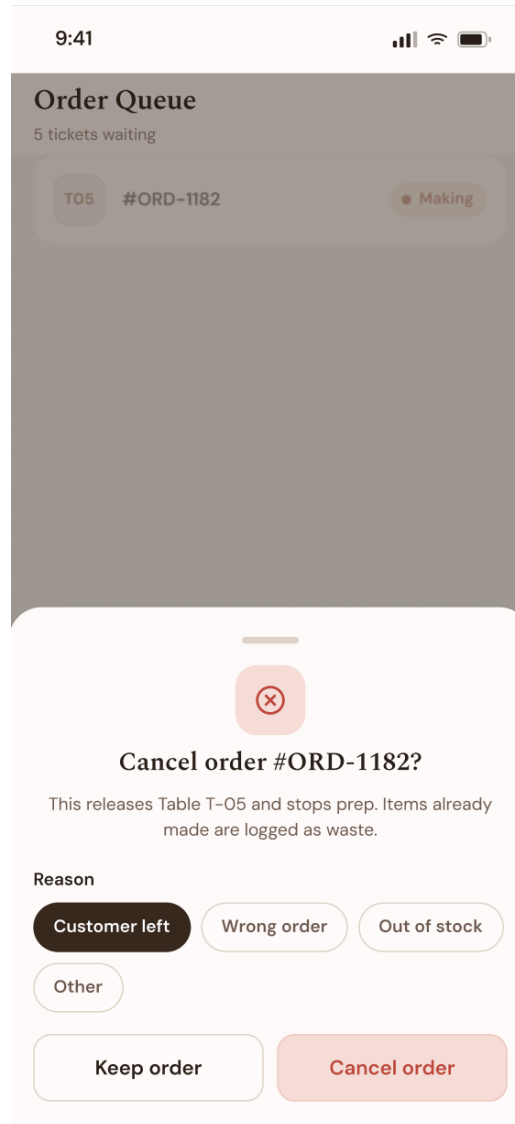


Table 3-5: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Close (X)	Button			Dismisses the dialog without cancelling the order.
2	Confirmation Message	Label			Displays the cancel

					confirmation text including the order ID and impact (e.g., releases the table and logs prepared items as waste).
3	Reason	Radio Button Group	Yes		Selects the cancellation reason (Customer left, Wrong order, Out of stock, Other).
4	Keep Order	Button			Closes the dialog and keeps the order unchanged.
5	Cancel Order	Button			Confirms cancellation and sets the order status to Cancelled.

3.5.3.2 Use Case Description

Use Case ID	UC08		Use Case Name		Cancel Order
Author	Project Team	Version	1.0	Date	12/06/2026
Actor		Cashier			
Description		The cashier cancels an open order.			
Precondition		An order with status Open exists.			
Trigger		Cashier selects Cancel Order.			
Post-Condition		The order status becomes Cancelled.			
Main flows					
Step		Actor		Action	
1		Cashier		Open an order, select Cancel.	

2	CSMS	Show confirmation MSG06.
3	Cashier	Confirm.
4	CSMS	Set status to Cancelled.
Alternative flows		
ID	Action	
AT1 / 2.1	Order already Paid → cannot cancel; corrections via refund/new order (BR-07).	

3.5.3 *FUNC-05-Order Management/ UC9: Assign Table*

3.5.3.1 Screen Mock-up

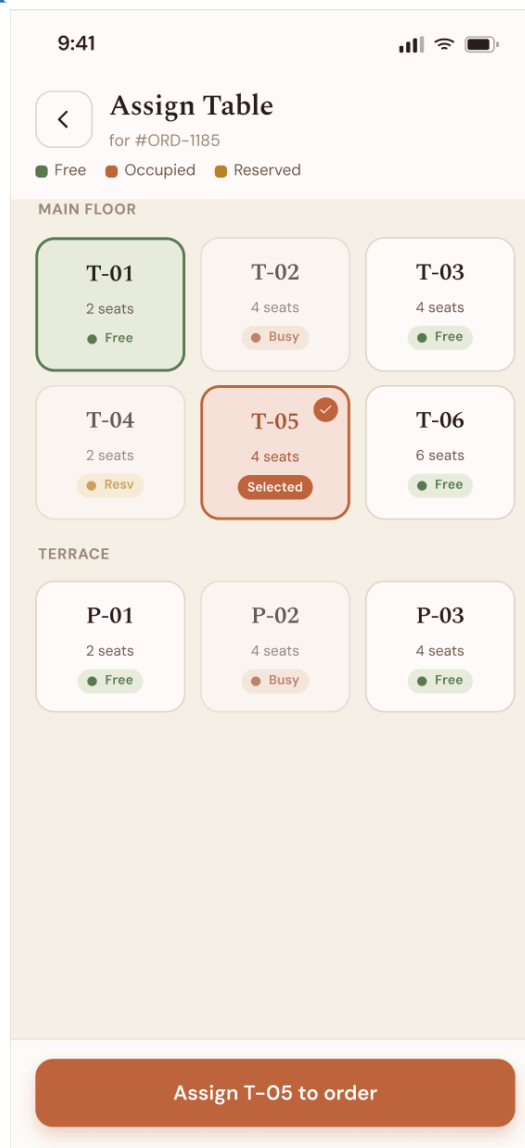


Table 3-6: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Back	Button			Returns to the previous screen without assigning a table.
2	Screen Title	Label			Displays the screen title and the order ID being assigned (e.g., for #ORD-1185).
3	Status Legend	Label			Shows the colour legend for table status (Free, Occupied, Reserved).
4	Floor Section	Label			Groups tables by area (Main Floor, Terrace).
5	Table Card	Container	No		Displays a table with its code, seat count and status; a free table can be tapped to select it (e.g., T-05).
6	Assign Button	Button			Links the selected table to the order and sets the table status to Occupied (e.g., Assign T-05 to order).

3.5.3.2 Use Case Description

Use Case ID	UC09		Use Case Name		Assign Table
Author	Project Team	Version	1.0	Date	12/06/2026
Actor		Cashier			
Description		The cashier assigns an order to a table or marks it takeaway. Extends Create Order.			
Precondition		An order is being created/edited; tables exist.			
Trigger		Cashier chooses a table during order creation.			
Post-Condition		The order is linked to a table; table status becomes Occupied.			
Extends		UC06 (Create Order)			
Main flows					
Step		Actor		Action	
1		Cashier		Select a free table (or Takeaway).	
2		CSMS		Link the order to the table, set table to Occupied.	
Alternative flows					
ID		Action			
AT1 / 1.1		Takeaway selected → TableID null.			

3.6 Product Management

3.6.1 Func 09- Manage Product UC13

3.6.1.1 Screen Mockup

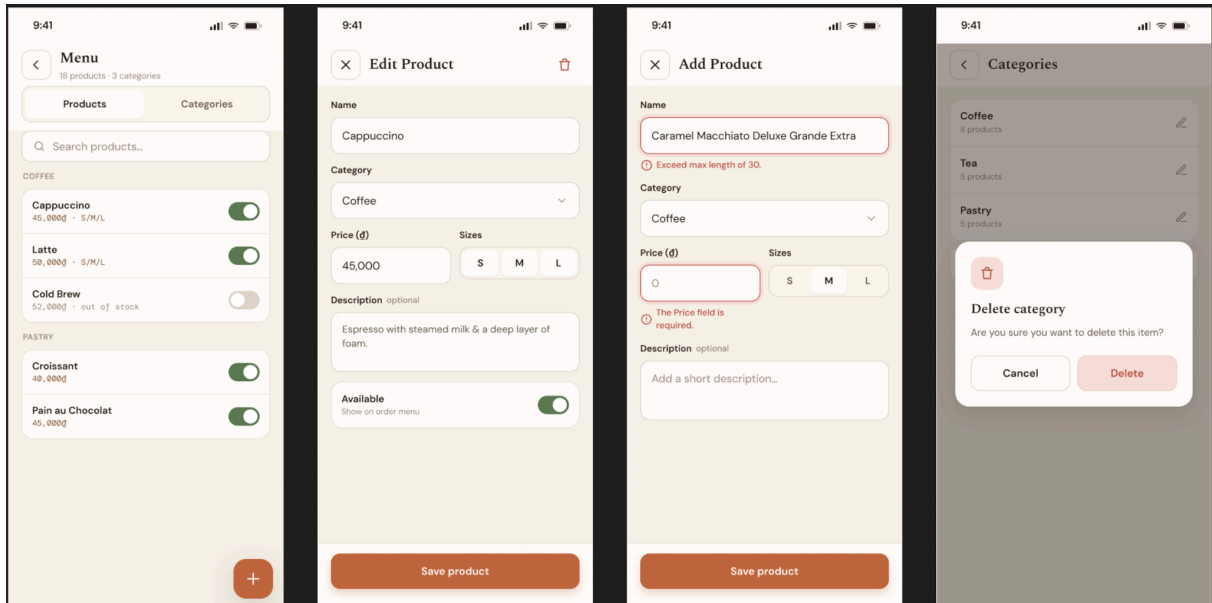


Table 3-3: Screen Definition — View Product

#	Field Name	Type	Mandatory	Max Length	Description
1	Back Button	Button	No		Navigate back to the previous dashboard screen.
2	Screen Title	Label	No	50	Display the static screen title “Menu”.
3	Menu Summary	Label	No	100	Display the short summary of items (e.g., “18 products · 3 categories”).

4	Products Tab	Tab / Button	No		Switch the list view to display all products (Active state).
5	Categories Tab	Tab / Button	No		Switch the list view to display product categories.
6	Search products Input	Text Input	No	100	Allow searching products by typing keywords.
7	Category Header	Label	No	50	Display the category name grouping the products (e.g., “COFFEE”, “PASTRY”).
8	Product Item Row	Container	No		Display product details including Name, Price, and available Sizes (or status like “out of stock”). Clicking the row triggers the Edit view.
9	Product Status Toggle	Toggle Switch	Yes		Quick enable or disable the product availability. Green indicates active status.

10	Add Product Button	Button	No		Floating Action Button (FAB) with a “+” icon to navigate to the Add Product screen.
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Table 3-4: Screen Definition — Edit Product

#	Field Name	Type	Mandatory	Max Length	Description
1	Close (X)	Button	No		Cancel the current operation and navigate back to the Menu Management list screen.
2	Screen Title	Label	No	50	Display the static screen title “Edit Product”.
3	Delete Icon	Button	No		Trigger the delete confirmation popup (MSG06) for this specific product.
4	Name Label	Label	No	30	Display the text label “Name”.
5	Name Input	Text Input	Yes	30	Enter or edit the product name. Default text: “Cappuccino”.

6	Category Label	Label	No	50	Display the text label “Category”.
7	Category Dropdown	Dropdown	Yes		Select a product category from the list. Current value: “Coffee”.
8	Price (₫) Label	Label	No	20	Display the text label “Price (₫)”.
9	Price Input	Text Input	Yes	15	Enter the price of the product. Displays value with comma separating thousands (e.g., “45,000”).
10	Sizes Label	Label	No	20	Display the text label “Sizes”.
11	Size Option S	Button / Checkbox	Yes		Toggle selection for size “S”.
12	Size Option M	Button / Checkbox	Yes		Toggle selection for size “M”.
13	Size Option L	Button / Checkbox	Yes		Toggle selection for size “L”.

14	Description Label	Label	No	30	Display the text label “Description” with optional hint.
15	Description Input	Text Input	No	255	Enter detailed product notes. Default: “Espresso with steamed milk & a deep layer of foam.”.
16	Available Setting Row	Label	No	50	Display setting row labels “Available” and description “Show on order menu”.
17	Available Toggle	Toggle Switch	Yes		Enable or disable the product availability. Default: Active (Green).
18	Save product	Button	No		Validate all input fields, save the changes, and show success message MSG05.

Table 3-5: Screen Definition — Add Product Validation (Màn hình 3)

#	Field Name	Type	Mandatory	Max Length	Description

1	Close (X)	Button	No		Cancel the product creation process and navigate back to the list screen.
2	Screen Title	Label	No	50	Display the static screen title “Add Product”.
3	Name Label	Label	No	30	Display the text label “Name”.
4	Name Input	Text Input	Yes	30	Input a new product name. Border turns red on error. Current input: “Caramel Macchiato Deluxe Grande Extra”.
5	Name Error Message	Label	No	50	Inline validation error text: “Exceed max length of 30.”.
6	Category Label	Label	No	50	Display the text label “Category”.
7	Category Dropdown	Dropdown	Yes		Select a product category from the list. Default: “Coffee”.

8	Price (₫) Label	Label	No	20	Display the text label “Price (₫)”.
9	Price Input	Text Input	Yes	15	Input the pricing value. Border turns red on empty field error. Current text: “0”.
10	Price Error Message	Label	No	50	Inline validation error text: “The Price field is required.”.
11	Sizes Label	Label	No	20	Display the text label “Sizes”.
12	Size S / M / L	Button Group	Yes		Select the supported sizes for the new product. Size “M” is active in the image.
13	Description Label	Label	No	30	Display the text label “Description optional”.
14	Description Placeholder	Text Input	No	255	Display helper placeholder text: “Add a short description...”.

15	Save product	Button	No		Execute field validations. Block operation until validation errors are resolved.
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Table 3-6: Screen Definition — Delete Product

#	Field Name	Type	Mandatory	Max Length	Description
1	Back Button	Button	No		Navigate back to the previous screen.
2	Screen Title	Label	No	50	Display the static screen title “Categories”.
3	Category List Items	Label	No	50	Display list elements containing category name and product amount count (e.g., “Coffee, 8 products”).
4	Edit Category Icon	Button	No		Trigger editing the selected category item line.
5	Dialog Overlay Background	Container	No		Darkened background blocking interactions with the background categories screen.

6	Dialog Alert Trash Icon	Icon	No		Red alert decorative trash icon at the top left of the dialog form.
7	Dialog Box Title	Label	No	50	Display confirmation card header text “Delete category”.
8	Delete Confirmation Message	Label	No	100	Display the message “Are you sure you want to delete this item?”.
9	Cancel	Button	No		Close delete confirmation dialog without modifications.
10	Delete	Button	No		Confirm deactivating/deleting the category, trigger success workflow, and close the dialog

3.6.1.2 Use case description

Use Case ID	UC13		Use Case Name		Manage Product (Add / Edit / Delete / View)
Author	Project Team	Version	1.0	Date	12/06/2026
Actor		Manager			
Description		The manager adds, edits, deletes or views products (name, category, price, size, availability, description).			
Precondition		Manager is logged in; at least one category exists (for Add).			
Trigger		Manager opens Menu Management → Product Details.			
Post-Condition		The product catalog is updated.			

Main flows		
Step	Actor	Action
1	Manager	Open Product Details, choose Add/Edit/Delete/View.
2	CSMS	Display the product form or details.
3	Manager	Enter/change values and Save (or confirm Delete).
4	CSMS	Validate and save (MSG05) or delete (MSG06 → MSG07).
Alternative flows		
ID	Action	
AT1 / 3.1	Required field empty → MSG02.	
AT2 / 3.2	Over max length → MSG08.	
Business Rules	BR-04, BR-05	

3.7 Inventory Management

3.7.1 UC17 View Inventory Report

3.7.1.1 Screen Mockup

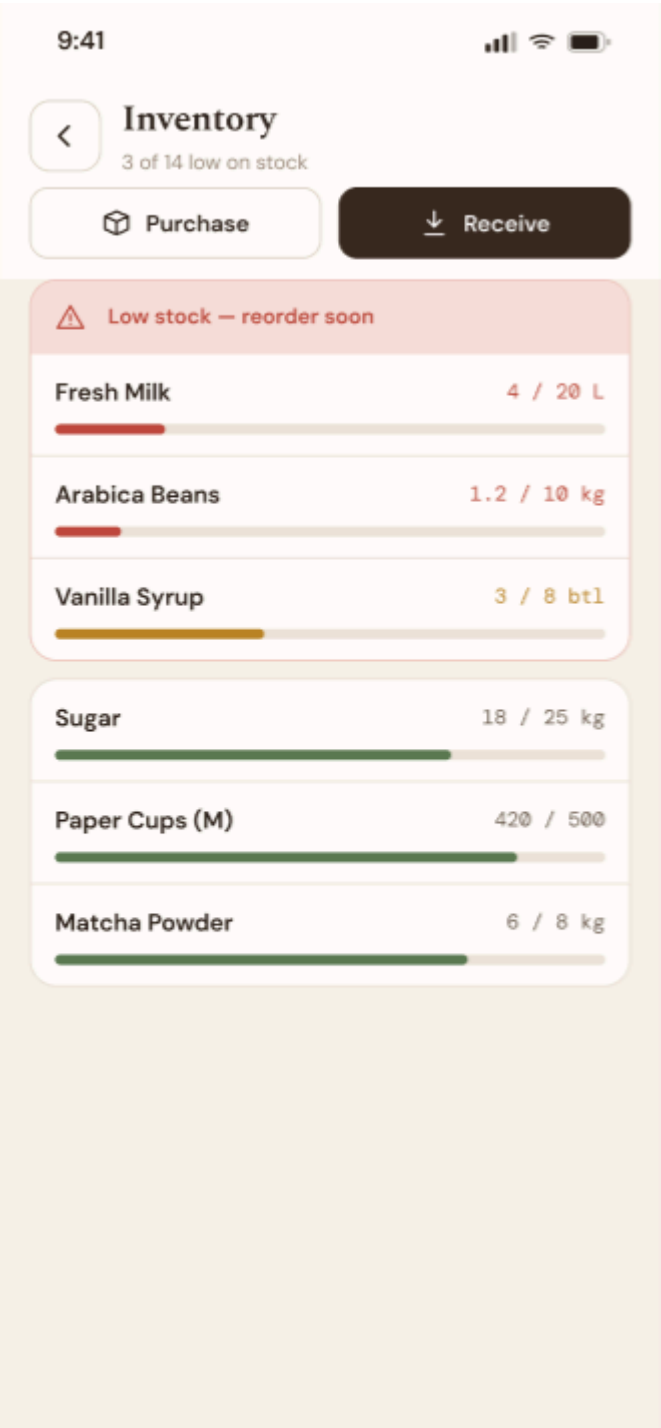


Table 3-7: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Purchase	Button	No		Opens the screen to create a new Purchase Order.
2	Receive	Button	No	50	Opens the Stock-In screen for receiving goods.
3	Low stock alert	Label	No	100	Display the short summary of items (e.g., “18 products · 3 categories”).
4	Ingredient list	List	No		Displays a warning (in red) for ingredients falling below the reorder threshold.

3.7.1.2 Use case description

Use Case ID	UC17		Use Case Name		View Inventory Report
Author	Project Team	Version	1.0	Date	12/06/2026
Actor		Manager			
Description		The manager reviews current stock levels and low-stock items.			
Precondition		Manager is logged in.			
Trigger		Manager opens the Inventory report.			
Post-Condition		The report is displayed.			
Main flows					
Step		Actor		Action	

1	Manager	Open the Inventory report.
2	CSMS	Display stock levels with low-stock items highlighted.

3.7.2 Func 08- Purchase Order

3.7.2.1 Screen Mock-up

9:41

<

Purchase Order

● Draft

#PO-0461 - Draft

Supplier

Saigon Coffee Supply Co. ▾

ITEMS

Arabica Beans 10 kg × 220,000đ	2,200,000đ
Fresh Milk 40 L × 28,000đ	1,120,000đ
Vanilla Syrup 12 btl × 95,000đ	1,140,000đ

+ Add line item

Total4,460,000đ

Save

Submit order

#	Field Name	Type	Mandatory	Max Length	Description
1	Supplier	Dropdown	Yes	-	Selects the ingredient supplier.
2	Items list	List	-	-	Displays the list of ordered ingredients, quantity, unit price, and line total.
3	Add line item	Button	-	-	Adds a new ingredient to the purchase order.
4	Total	Label	-	-	Displays the total cost of the purchase order.
5	Save	Button	-	-	Saves the purchase order as a Draft.
6	Submit order	Button	-	-	Confirms and submits the purchase order to the supplier.

3.7.3 UC16 Create Stock-In

3.7.3.1 Screen Mockup

9:41

<

Stock-In

Receive against #PO-0461

Saigon Coffee Supply Co.

PO-0461 · ordered Jun 4

RECEIVED QUANTITIES

Arabica Beans

Ordered 10 kg

—

10

+

Fresh Milk

Ordered 40 L

—

38

+

●

2 short of order

Vanilla Syrup

Ordered 12 btl

—

12

+

Confirm goods receipt

#	Field Name	Type	Mandatory	Max Length	Description
1	PO Info	Label			Displays the supplier's name and the referenced Purchase Order ID.
2	Received Quantities	List			Lists the ordered ingredients and the expected quantities.
3	Quantity Input	Numeric Input	Yes		Allows adjusting the actual received quantity using (- / +) buttons.
4	Shortage alert	Label			Displays an alert if the actual received quantity is less than ordered (e.g., "2 short of order").
5	Confirm goods receipt	Button			Confirms the receipt of goods and updates the stock on hand.

3.7.3.2 Use case description

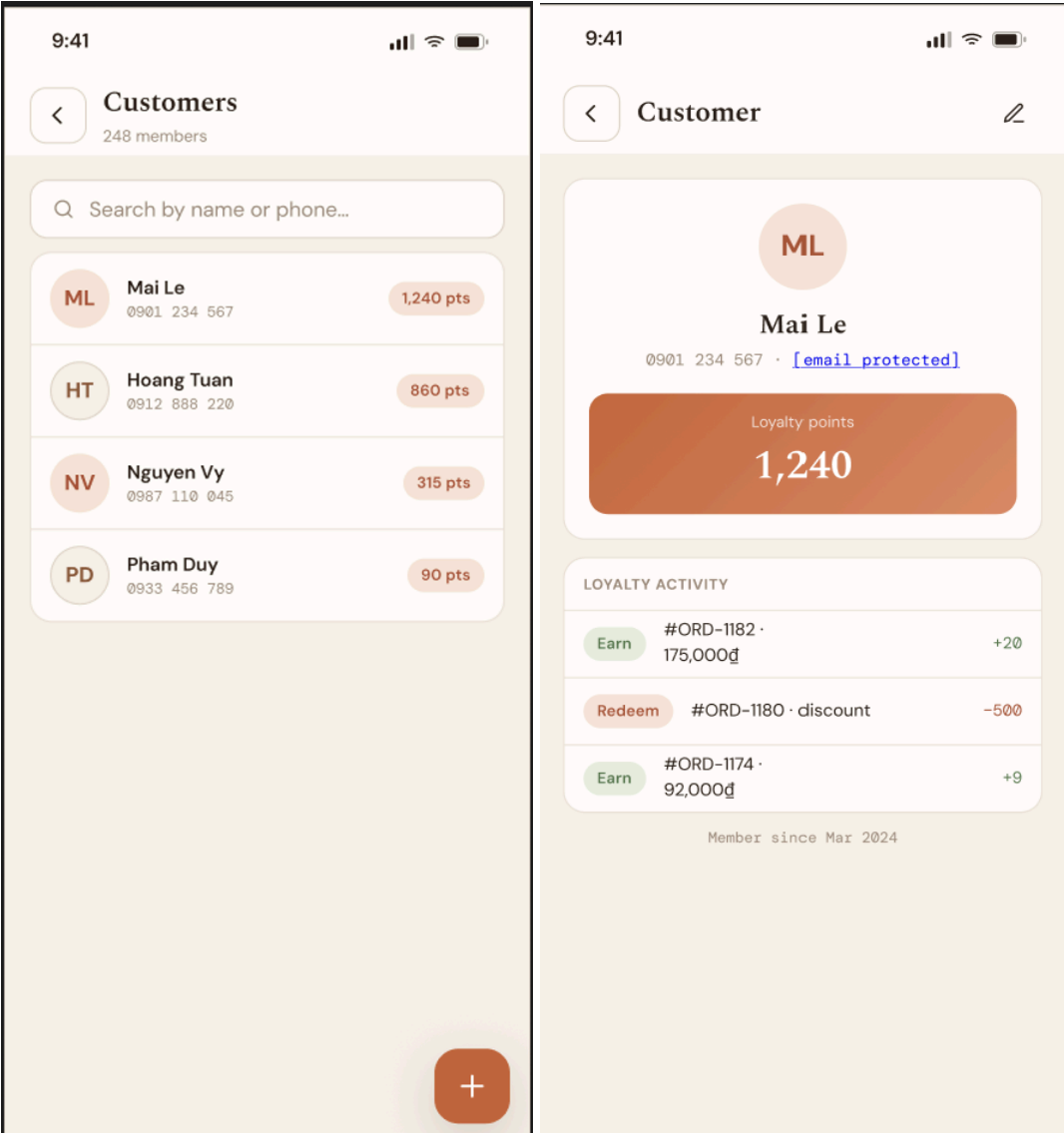
Use Case ID	UC16		Use Case Name		View Inventory Report
Author	Project Team	Version	1.0	Date	12/06/2026
Actor		Manager			
Description		Records received goods against a purchase order			

Precondition	A purchase order exists.	
Trigger	Manager opens Stock-In and selects a purchase order.	
Post-Condition	Stock-In records are saved; ingredient QuantityOnHand increases.	
Main flows		
Step	Actor	Action
1	Manager	Open Stock-In, select a purchase order.
2	CSMS	List the ordered ingredients.

3.8 Customer Management

3.8.1 UC19: Manage Customer

3.8.1.1 Screen Mock-up



#	Field Name	Type	Mandatory	Max Length	Description

1	Back Button	Button	No		Navigate back to the previous dashboard screen.
2	Screen Title	Label	No	50	Display the static screen title “Customers”.
3	Members Summary	Label	No	100	Display the total count of registered members (e.g., “248 members”).
4	Search Input	Text Input	No	100	Allow searching customers by entering name or phone number. Placeholder: “Search by name or phone...”.
5	Customer List Container	Container	No		Scrollable panel containing the row elements for all registered customers.
6	Customer Avatar	Image / Icon	No		Circular badge displaying the customer's initials (e.g., “ML”, “HT”, “NV”, “PD”).
7	Customer Name	Label	No	50	Display the full name of the customer (e.g., “Mai Le”, “Hoang Tuan”).
8	Phone Number	Label	No	20	Display the registered mobile number format (e.g., “0901 234 567”).

9	Loyalty Points Badge	Label	No	20	Display the total accumulated loyalty points balance (e.g., “1,240 pts”).
10	Customer Item Row	Container	No		Trigger navigation to the Customer Detail view when clicked.
11	Add Customer Button	Button	No		Floating Action Button (FAB) with a “+” icon to navigate to the Add Customer screen.

#	Field Name	Type	Mandatory	Max Length	Description
1	Back Button	Button	No		Navigate back to the Customer Management list screen.
2	Screen Title	Label	No	50	Display the static screen title “Customer”.
3	Edit Icon	Button	No		Navigate to the Edit Customer Form screen.
4	Customer Profile Badge	Image / Icon	No		Large circular badge displaying the customer's initials (“ML”).

5	Customer Name	Label	No	50	Display the detailed profile full name (“Mai Le”).
6	Contact Information	Label	No	150	Display the phone number and protected email address string.
7	Loyalty Points Card	Container	No		Highlight card section component with background fill for loyalty points display.
8	Loyalty Points Value	Label	No	20	Display the aggregate points sum value text (“1,240”).
9	Loyalty Activity Header	Label	No	50	Display section grouping header title “LOYALTY ACTIVITY”.
10	Activity History List	Container	No		List container displaying current point transactional operations.
11	Activity Type Tag	Label	No	20	Display operation type badge flag (e.g., “Earn” in green or “Redeem” in orange).

12	Activity Detail	Label	No	100	Display reference order ID and metrics (e.g., “#ORD-1182 · 175,000d” or “#ORD-1180 · discount”).
13	Activity Points Delta	Label	No	20	Display the numeric change value with math operators (e.g., “+20” or “-500”).
14	Membership Timestamp	Label	No	50	Display chronological registration summary text

3.8.1.2 Use case description

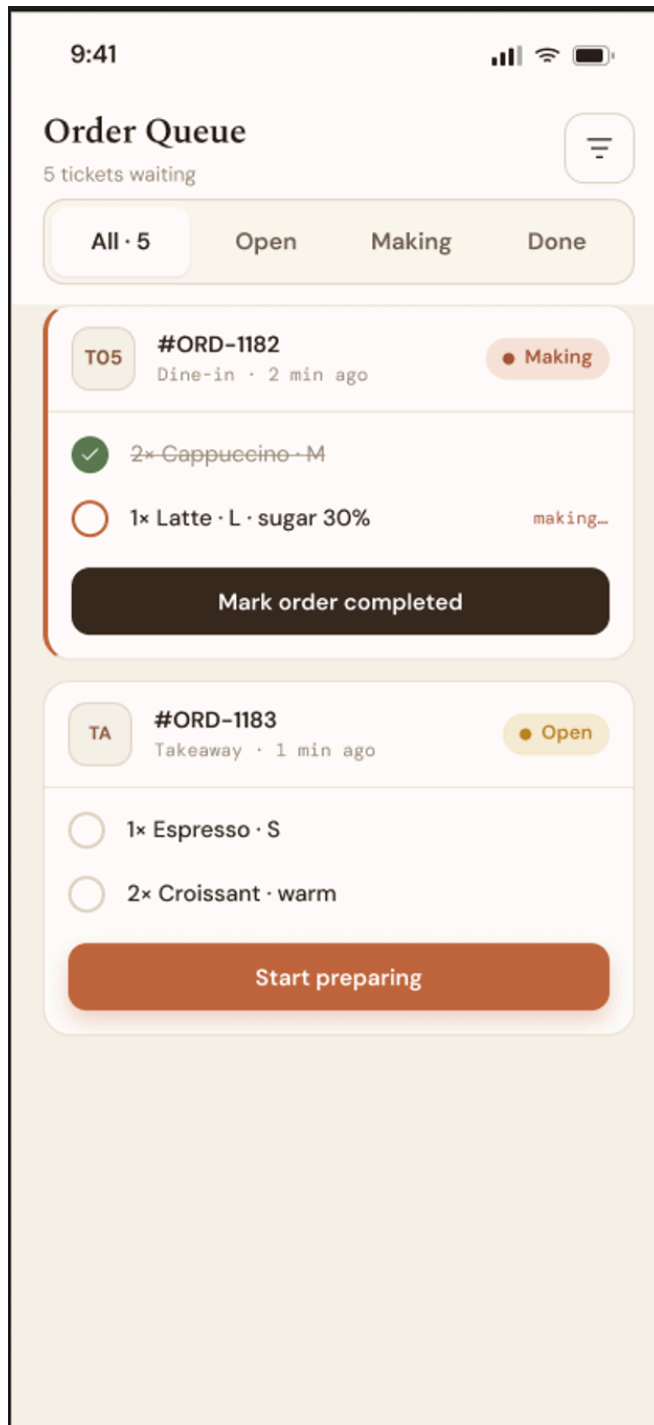
Use Case ID	UC19		Use Case Name		Manage Customer (Add / Edit / Delete / View)
Author	Project Team	Version	1.0	Date	12/06/2026
Actor		Manager			
Description		The manager adds, edits, deletes or views customer records and loyalty balance.			
Precondition		Manager is logged in.			
Trigger		Manager opens Customer Management.			
Post-Condition		The customer list is updated.			
Main flows					
Step		Actor		Action	
1		Manager		Open Customer Management, choose Add/Edit/Delete/View.	
2		CSMS		Display the form or details.	

3	Manager	Enter/change values and Save (or confirm Delete).
4	CSMS	Apply the change (MSG05 / MSG06 → MSG07).
Alternative flows		
ID	Action	
AT1 / 3.1	Required field empty → MSG02.	
AT2 / 3.2	Over max length → MSG08.	

3.9 Order Queue

3.9.1 UC11: View Order Queue

3.9.1.1 Screen Mock-up



#	Field Name	Type	Mandatory	Max Length	Description
1	Screen Title	Label	No	50	Display the static screen title “Order Queue”.
2	Queue Summary	Label	No	100	Display the short summary of waiting tickets (e.g., “5 tickets waiting”).
3	Filter Settings Button	Button	No		Icon button to open advanced filtering configurations for the order list.
4	All Tab	Tab / Button	No		Filter and display all tickets in the queue (Active state, showing “All · 5”).
5	Open Tab	Tab / Button	No		Filter and display only tickets with an Open status.
6	Making Tab	Tab / Button	No		Filter and display only tickets currently in preparation.
7	Done Tab	Tab / Button	No		Filter and display only completed tickets.

8	Ticket Card Container	Container	No		Card module displaying comprehensive details for a single order ticket.
9	Table / Service Badge	Label	No	20	Display the location identification (e.g., “T05” for Table 5 or “TA” for Takeaway).
10	Order ID Header	Label	No	30	Display the unique system reference code (e.g., “#ORD-1182”, “#ORD-1183”).
11	Service Type & Time	Label	No	100	Display fulfillment class and creation duration metadata (e.g., “Dine-in · 2 min ago” or “Takeaway · 1 min ago”).
12	Ticket Status Badge	Label	No	20	Colored indicator badge reflecting the processing phase (e.g., “Making” in orange or “Open” in yellow).

13	Item Checkbox Indicator	Checkbox / Button	No		Interactive circle indicator representing item state. Changes to a green checkmark when an individual product item is completed.
14	Order Item Line	Label	No	150	Display the requested product quantity, name, and options (e.g., “2× Cappuccino · M” with strikethrough text if done, or “1× Latte · L · sugar 30%”).
15	Individual Item Status	Label	No	20	Minor status alert text aligned to the right of an item (e.g., “making...” in orange).
16	Mark order completed	Button	No		Action button available on tickets in preparation to change the entire order status to Completed.

17	Start preparing	Button	No		Action button available on unfulfilled tickets to change the entire order status to In-progress.
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3.9.1.2 Use case description

Use Case ID	UC11		Use Case Name		View Order Queue
Author	Project Team	Version	1.0	Date	12/06/2026
Actor		Barista, Cashier, Manager			
Description		The user views open orders and the preparation status of items.			
Precondition		User is logged in.			
Trigger		User opens the Order Queue.			
Post-Condition		The current queue is displayed.			
Extended by		UC12 (Update Item Prep Status)			
Main flows					
Step		Actor		Action	
1		User		Open the Order Queue.	
2		CSMS		Display open orders and per-item prep status.	

3.10 Report

3.10.1 UC20: View Sales Report

3.10.1.1 Screen Mock-up

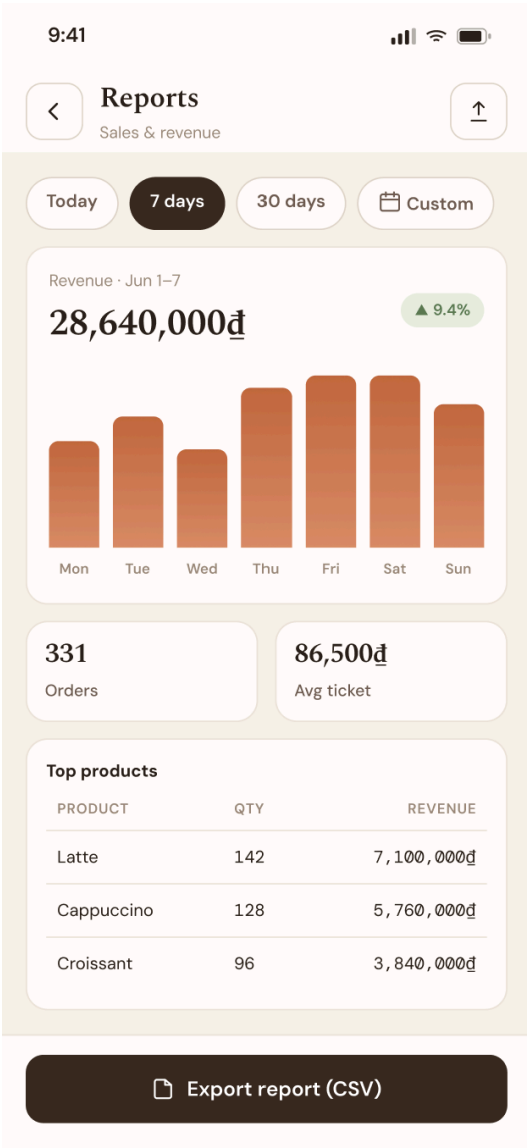


Table 3-11: Screen Definition

#	Field Name	Type	Mandatory	Max Length	Description
1	Back	Button			Returns to the previous screen.
2	Export Icon	Button			Opens export options for the current report (shortcut in the header).

3	Date Range Tabs	Tab / Button	No		Selects the reporting period (Today, 7 days, 30 days, Custom).
4	Revenue Summary	Label			Displays total revenue for the selected period with a comparison indicator (e.g., 28,640,000đ, ▲9.4%).
5	Revenue Chart	Container			Bar chart of daily revenue across the selected period (Mon–Sun).
6	Stat Cards	Label			Shows summary metrics for the period, such as number of orders and average ticket.
7	Top Products	Container			Lists best-selling products with quantity (QTY) and revenue columns.
8	Export Report (CSV)	Button			Exports the current report as a CSV file.

3.10.1.2 Use case description

Use Case ID	UC20		Use Case Name		View Sales Report
Author	Project Team	Version	1.0	Date	12/06/2026

Actor	Manager	
Description	The manager views sales and revenue reports filtered by date range.	
Precondition	Manager is logged in.	
Trigger	Manager opens Reports and selects a date range.	
Post-Condition	The report is displayed.	
Extended by	UC21 (Export Report)	
Main flows		
Step	Actor	Action
1	Manager	Open Reports, select a date range.
2	CSMS	Display sales/revenue and best-selling products for the range.
Alternative flows		
ID	Action	
AT1 / 2.1	No data in range → MSG01.	

Use Case ID	UC21		Use Case Name		Export Report
Author	Project Team	Version	1.0	Date	12/06/2026
Actor		Manager			
Description		The manager exports the currently viewed report. Includes View Sales Report.			
Precondition		A report is displayed.			
Trigger		Manager selects Export.			
Post-Condition		A report file is generated and downloaded.			
Includes		UC20 (View Sales Report)			
Main flows					
Step		Actor		Action	
1		Manager		Select Export on the report.	
2		CSMS		Generate and download the report file.	

4. Non-Functional Requirements

4.1 External Interfaces

User Interfaces: a Flutter-based mobile application on Android and iOS tablets/phones, optimized for fast counter order entry.

Hardware Interfaces: a receipt/thermal printer for receipts and kitchen tickets; an optional barcode scanner for product lookup.

Software Interfaces: an external payment gateway (REST API over HTTPS) for card and e-wallet payments.

Communication Interfaces: the app communicates with the backend over HTTPS via the cafe's Wi-Fi/internet; gateway calls also use HTTPS.

4.2 Quality Attributes

4.2.1 Usability

A new cashier should learn to create and complete an order within 30 minutes of training. A typical order of up to 5 items should be enterable in under 60 seconds. Clear labels, large touch targets and consistent navigation keep frequent tasks (create order, take payment) within two taps of the Dashboard.

4.2.2 Reliability

Availability $\geq 99.5\%$ during operating hours. MTTR ≤ 30 minutes. The system auto-recovers committed order and payment data after a restart. Order and payment totals calculated with 100% accuracy. No critical defect (loss of committed order/payment data, or inability to take payment) remains in a production release.

4.2.3 Performance

Screen transitions and product searches respond within 1 second on average, 3 seconds at peak. Saving an order or processing a payment completes within 2 seconds on average. Supports ≥ 20 concurrent users and ≥ 60 orders/hour at peak without degradation. Stores ≥ 2 years of order/sales history while maintaining the above response times.

4.2.4 Security

Passwords stored hashed. All transmission of payment and personal data uses ≥ 256 -bit TLS encryption. Role-based authorization per Section 3.1.3. Audit log of login, payment and data-modification events.

5. Requirement Appendix

5.1 Business Rules

ID	Rule Definition
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BR-01	An order must contain at least one product before it can be saved or paid.
BR-02	Order total = sum of (product price × quantity) for all items, minus any applicable discount.
BR-03	All items in a single order must be paid using a single payment method.
BR-04	A product cannot be ordered when marked unavailable or out of stock.
BR-05	Only a Manager or Administrator can create, modify or deactivate products and prices.
BR-06	A discount may not exceed 50% of the order subtotal unless approved by a Manager.
BR-07	Once an order is paid it cannot be edited; corrections via refund or new order.
BR-08	Ingredient stock is reduced automatically when a paid order contains products that use it.
BR-09	Network transmissions involving payment or personal data require at least 256-bit encryption.
BR-10	An account is locked for 15 minutes after 5 consecutive failed login attempts.
BR-11	Loyalty points accrue only on completed payments; redeemed points cannot exceed the customer's balance. [EDIT — confirm loyalty rule]
BR-12	Stock-In quantity must reference an existing Purchase Order line. [EDIT — confirm]

5.2 Common Requirements

ID	Requirement
CR-01	All mandatory fields shall be marked with a visual indicator (*) and validated before data is saved.
CR-02	All date and time values shall use the system date/time and be displayed in the format DD/MM/YYYY HH:mm.

CR-03	The system shall display a success, warning, or error message after every create, update, delete, login, or payment action.
CR-04	The system shall prevent duplicate usernames within the User entity.
CR-05	The system shall automatically record CreatedAt and UpdatedAt timestamps for all transactional records.
CR-06	Only authenticated users may access system functions. Unauthorized users shall be redirected to the Login screen.
CR-07	Users shall only access functions permitted by their assigned role according to the authorization matrix in Section 3.1.3.
CR-08	Data entered into text fields shall be trimmed of leading and trailing spaces before validation.
CR-09	The system shall require confirmation before performing irreversible actions such as deleting records or cancelling orders.
CR-10	All monetary values shall be displayed in Vietnamese Dong (VND) using thousand separators.
CR-11	The system shall maintain an audit log for login, logout, payment, and administrative actions.
CR-12	The system shall automatically log out inactive users after 30 minutes of inactivity.
CR-13	The system shall support searching and filtering lists where applicable (users, customers, products, orders).
CR-14	All screens shall provide a consistent navigation mechanism and a way to return to the previous screen.
CR-15	The system shall validate all user input before sending data to the backend services.

5.3 Application Messages List

#	Code	Type	Context	Content
1	MSG01	In line	No search result	No search results.
2	MSG02	In red, under text box	Required field empty	The {field} field is required.
3	MSG03	Toast	Order created	Order created successfully.
4	MSG04	Toast	Payment completed	Payment completed successfully.
5	MSG05	Toast	Product saved	Product saved successfully.
6	MSG06	Confirmation dialog	Confirm deletion	Are you sure you want to delete this item?
7	MSG07	Toast	Record deleted	Item deleted successfully.
8	MSG08	In red, under text box	Length exceeds max	Exceed max length of {max_length}.
9	MSG09	In line	Wrong username/password	Incorrect user name or password. Please check again.

5.4 Other Requirements...

There are no additional requirements at this time. Any further requirements identified during development will be documented here and reflected in the Record of Changes.